

Exploring Lombardy Through Data

Pollution, Weather, Mobility and Tourism Data from 2023

Predictive Analytics Project - 101801

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1 Introduction

This project analyzes air quality, weather, traffic, and tourism across Lombardy in 2023. The data comes from ARPA Lombardia, the Municipality of Milan, and ISTAT.

The year 2023 was chosen because the traffic dataset only covers that period. Restricting all sources to the same window keeps things consistent.

Four data domains feed into the analysis:

- **Air and Weather:** hourly and daily sensor readings for pollutants and meteorological variables across Lombardy (ARPA Lombardia).
- **Traffic:** individual vehicle transit records at Milan's Area C restricted zone (Municipality of Milan).
- **Tourism:** monthly overnight stays per municipality (ISTAT).
- **Territory:** population counts, municipal statistics, and geographic boundaries (ISTAT).
- **Geospatial:** municipality boundaries in GeoJSON format (OpenStreetMap).

The work proceeds in two stages:

- A Data Warehouse is designed and filled through an ETL pipeline into a PostgreSQL database.
- Then, **SQL Queries**, **Association Rules**, and **Machine Learning** are applied to answer the business questions laid out in the next section.

2 Business Questions

How can air quality data across municipalities be analyzed to support environmental and public health decision-making?

- Which urbanization levels show the highest pollutant concentrations?
- Are there municipalities that repeatedly experience multi-day episodes of pollution above health-based thresholds?
- Can municipalities be grouped based on their pollution profile?
- Which municipalities are most polluted with particulate matter?

How do weather variables relate to pollutant concentrations across municipalities?

- Which weather variables are most strongly correlated with pollutant concentrations?
- Which combinations of weather conditions frequently co-occur with pollution levels?

How can vehicle access data be analyzed to model patterns, forecast trends, and detect anomalies in support of mobility planning?

- What is the composition of the fleet of vehicles accesses and how is it distributed?
- Can daily vehicle accesses be predicted?
- Which days show anomalous traffic volumes?

Is there a measurable association between human activity intensity and air pollutant levels?

- Do high traffic days co-occur with elevated pollutant levels?
- Does high tourism pressure co-occur with higher pollution levels?

3 Datasets

3.1 Lombardia Air Quality Sampling

Datasets Source

Hourly or daily air pollution readings from monitoring stations across Lombardy. Each station hosts one or more pollutant sensors.

Schema

Field	Description
idSensor	Sensor ID
Date	Date and time of the measurement
Value	Pollutant level (-999 means missing)
Status	Data status (VA = valid, NA = invalid)

Notes: All pollutants are measured hourly, except PM₁₀ and PM_{2.5} which are recorded as daily averages. Values of -999 indicate missing data.

3.2 Lombardia Air Quality Sensor and Station Registry

Datasets Source

Registry of all air monitoring sensors and their stations. Station information is denormalized one row per sensor.

Schema

Field	Description
idSensor	Sensor ID
SensorTypeName	Pollutant measured
Unit	Unit of measurement
idStation	Station ID
StationName	Station name
Altitude	Station height (m)
Municipality	Town
Province	Province
StartDate	Start activity date
StopDate	End activity date
Historical	Historical flag
UTM_North	UTM north coordinate
UTM_East	UTM east coordinate
Lng	Longitude
Lat	Latitude
Location	Coordinates

3.3 Lombardia Weather and Environmental Sampling

- Snow Depth Datasets Source
- Precipitation Datasets Source
- Temperature Datasets Source

- Humidity Datasets Source
- Solar Radiation Datasets Source
- Wind Speed Datasets Source
- Wind Direction Datasets Source
- Wind Gusts Datasets Source

ARPA Lombardia publishes one dataset per measurement type, but they all share the same schema. They were merged before ETL to keep things simple. Sampling frequencies differ by variable.

Variable	Unit	Frequency
Temperature	°C	Every 10 minutes
Humidity	%	Every 10 minutes
Precipitation	mm	Every 10 minutes
Solar radiation	W/m ²	Every 10 minutes
Wind speed	m/s	Every 10 minutes
Wind direction	°	Every 10 minutes
Wind gusts	m/s	Every 10 minutes
Water level	cm	Hourly
Snow depth	cm	Hourly

Schema

Field	Description
idSensor	Sensor ID
Date	Date and time of the measurement
Value	Measured value
Status	Data status

Notes: Values of -999 indicate missing data.

3.4 Lombardia Weather Station Registry

Datasets Source

Registry of hydro-nivo-meteorological stations and their sensors. One row per sensor, station information is denormalized.

Schema

Field	Description
idSensor	Sensor ID
MeasurementType	Type of measurement
Unit	Unit of measurement
idStation	Station ID
StationName	Name of the station
Province	Province
Altitude	Station height (m)
StartDate	Start activity date
StopDate	End activity date
Historical	Historical flag
UTM_North	UTM north coordinate
UTM_East	UTM east coordinate
Lng	Longitude
Lat	Latitude
Location	Coordinates

3.5 Milan Area C Gate Registry

Datasets Source

Geographic registry of the electronic gates delimiting Milan's Area C congestion zone.

Field	Description
id_amat	Gate ID
label	Gate name
LONG_X_4326	Longitude
LAT_Y_4326	Latitude
Location	Coordinates

3.6 Milan Area C Gate Ingress Sampling

Monthly files for 2023:

- January Dataset Source
- February Dataset Source
- March Dataset Source
- April Dataset Source
- May Dataset Source
- June Dataset Source
- July Dataset Source
- August Dataset Source
- September Dataset Source
- October Dataset Source
- November Dataset Source
- December Dataset Source

Vehicle transits detected by Area C gate cameras, released by the Municipality of Milan as monthly files. Same schema throughout, so they were merged before ETL ingestion.

Field	Description
dataora	Timestamp of the vehicle transit.
id_varco	Unique identifier of the transit gate.
esenti	Indicates if the vehicle was exempt.
moto	Indicates if the vehicle was a motorcycle.
residenti	Indicates if the vehicle owner was a resident.
veicoli_servizio	Indicates if the vehicle was a service vehicle.
categoria_euro	Euro emission standard of the vehicle.
tipologia_alimentazione	Fuel type of the vehicle.
categoria_veicolo	Category of the vehicle.
classe_areac	Area C vehicle class.
fap	Indicates if the vehicle has a particulate filter.
areac	Indicates if the vehicle enters or exits the area.
numero_transiti	Number of transits for this vehicle at this gate and time.

3.7 Milan Area C Decode Field

Datasets Source

Lookup table mapping coded values in the ingress dataset to human-readable descriptions.

Field	Description
fieldname	Name of the field
id_amat	Coded value
descrizione	Description

3.8 Tourist Overnight Stays

Datasets Source

Monthly tourism flows per municipality and accommodation type across Lombardy.

Field	Description
Anno	Reference year.
Provincia	Province name.
Codice ISTAT	ISTAT municipality code.
Comune	Municipality name.
Tipo struttura	Accommodation type.
Mese	Month name.
Arrivi	Number of guest arrivals.
Presenze	Total overnight stays.
Permanenza media	Average length of stay.

3.9 Municipalities Statistics

Datasets Source

Geographic and demographic statistics for all Italian municipalities, published by ISTAT.

Field	Description
Codice Regione	Region code.
Codice Istat del Comune (alfanumerico)	ISTAT municipality code.
Codice Istat del Comune (numerico)	ISTAT municipality code.
Denominazione (Italiana e straniera)	Municipality name.
Denominazione altra lingua	Name in another language.
Superficie territoriale (kmq) al 01/01/2024	Total area in km as of 01/01/2024.
Popolazione legale 2021 (31/12/2021)	Legal population as of 31/12/2021.
Popolazione residente al 31/12/2023	Resident population as of 31/12/2023.
Zona altimetrica	Altimetric zone code.
Altitudine del centro (metri)	Altitude of the municipal center in meters.
Comune litoraneo	Coastal municipality flag.
Comune isolano	Island municipality flag.
Zone costiere	Coastal zone flag.
Grado di urbanizzazione	Urbanization level code

3.10 Population Census

Datasets Source

Annual demographic counts for all Italian municipalities, published by ISTAT.

Schema

Field	Description
Codice comune	ISTAT municipality code.
Comune	Municipality name.
Popolazione censita al 31 dicembre - Totale	Total resident population at 31 December.

3.11 Municipalities Shape

Datasets Source

Municipality boundaries in GeoJSON format, provided by OpenStreetMap.

4 Data Warehouse Structure

4.1 Design

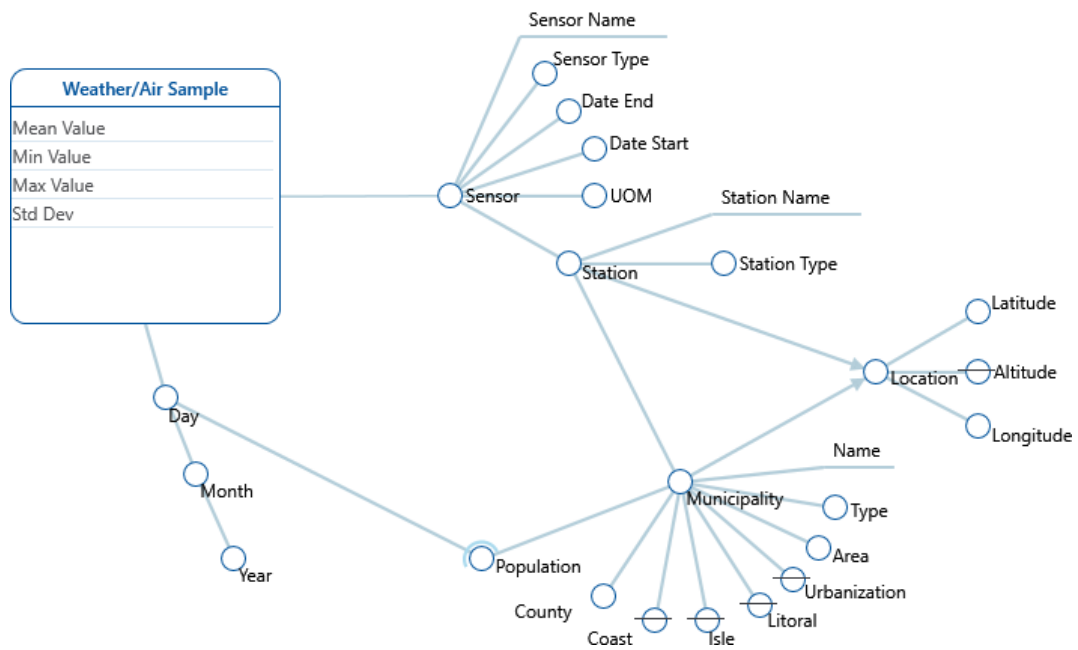


Figure 1: Weather and Air Quality Sample DFM

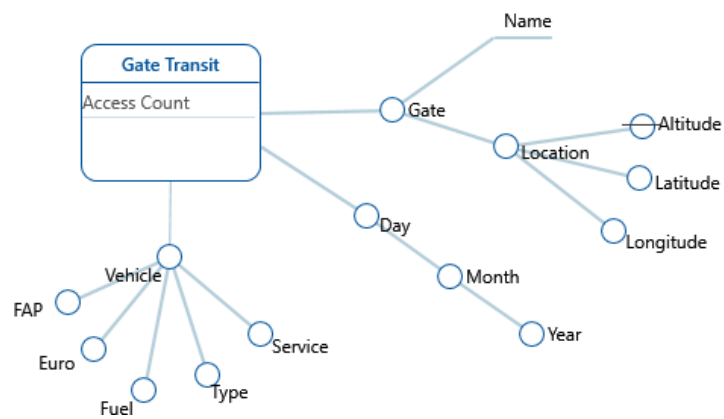


Figure 2: Gate Access DFM

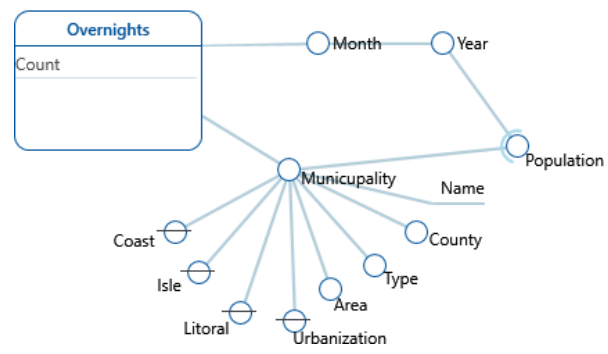
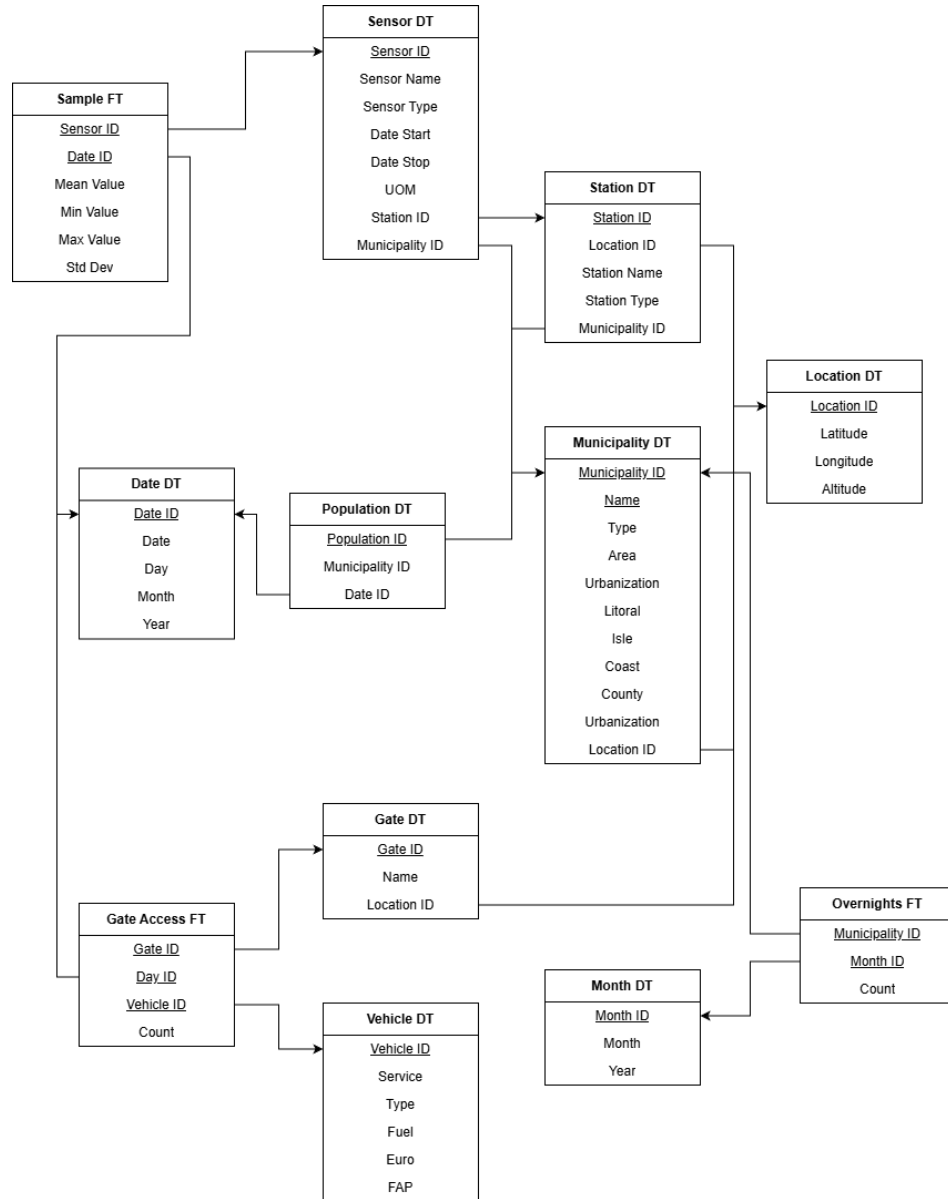


Figure 3: Overnights DFM

Schemas designed with Indyco.

4.2 Snowflake Schema

The three DFM models were merged into a single **snowflake schema** instead of three separate star schemas. Shared concepts: coordinates (**location_dt**), municipality metadata (**municipality_dt**), and date (**date_dt**) live in dedicated tables that multiple fact tables and dimensions reference. This cuts redundancy across **station_dt**, **gate_dt**, and **municipality_dt**, keeps natural hierarchies intact (sensor → station → municipality → location), and makes cross-fact joins possible, for instance linking **sample_ft** and **overnights_ft** through **municipality_dt** without duplicating data.



4.3 Fact Tables

4.3.1 Weather and Air Quality Sample Fact (`sample_ft`)

Daily aggregated measurements from a single sensor, either air quality or weather. One row per sensor per day. Time is given by `date_dt`; the sensor links through `sensor_dt` to its station and municipality.

Column	Type	Description
<code>sensor_id</code>	integer	FK with <code>sensor_dt</code>
<code>date_id</code>	integer	FK with <code>date_dt</code>
<code>mean_value</code>	double	Daily average measurement
<code>min_value</code>	double	Daily minimum
<code>max_value</code>	double	Daily maximum
<code>std_dev</code>	double	Daily standard deviation

4.3.2 Area C Gate Access Fact (`gate_access_ft`)

Number of transits for a given vehicle profile through a given Area C gate on a given day. One vehicle profile-gate-day combination.

Column	Type	Description
<code>gate_id</code>	integer	FK with <code>gate_dt</code>
<code>date_id</code>	integer	FK with <code>date_dt</code>
<code>vehicle_id</code>	integer	FK with <code>vehicle_dt</code>
<code>count</code>	integer	Number of transits

4.3.3 Tourism Overnights Fact (`overnights_ft`)

Total overnight stays in a municipality during a given month. One municipality-month combination.

Column	Type	Description
<code>municipality_id</code>	integer	FK with <code>municipality_dt</code>
<code>month_id</code>	integer	FK with <code>month_dt</code>
<code>count</code>	integer	Monthly overnight stays

4.4 Dimension Tables

4.4.1 Date Dimension (`date_dt`)

Temporal attributes for each daily measurement, breaking a full date into day, month, and year.

Column	Type	Description
<code>date_id</code>	integer	Primary key
<code>date</code>	date	Full date
<code>day</code>	smallint	Day of month
<code>month</code>	smallint	Month
<code>year</code>	smallint	Year

4.4.2 Month Dimension (month_dt)

Month and year, used only by `overnights_ft`. A separate `month_dt` was created instead of reusing `date_dt` for two reasons. First, tourism data is monthly there is no day-level detail. Linking `overnights_ft` to `date_dt` would force every overnight record onto an arbitrary day (say, the first of the month), adding false precision that does not exist in the source. Second, joining to `date_dt` would blow up each municipality-month row to as many as 31 rows, one per day, wasting storage for zero analytical gain. A dedicated `month_dt` keeps the fact at its natural grain, respects the snowflake semantics, and avoids pointless overhead.

Column	Type	Description
month_id	integer	Primary key
month	smallint	Month
year	smallint	Year

4.4.3 Location Dimension (location_dt)

Latitude, longitude, and altitude, shared by municipalities, monitoring stations, and Area C gates. Putting coordinates in one table removes duplication from `station_dt`, `gate_dt`, and `municipality_dt`.

Column	Type	Description
location_id	integer	Primary key
latitude	double precision	Latitude
longitude	double precision	Longitude
altitude	double precision	Altitude in meters (NULL for gates)

4.4.4 Municipality Dimension (municipality_dt)

Descriptive attributes for every Lombardy municipality: name, urbanization level, area, altimetric zone, coastal and island flags, and a pointer to geographic coordinates in `location_dt`.

Column	Type	Description
municipality_id	integer	Primary key
name	varchar(100)	Municipality name
urbanization	varchar(100)	Urbanization level label
area	double	Territory area
littoral	boolean	Coastal municipality flag
isle	boolean	Island municipality flag
coast	boolean	Coastal zone flag
altitude_zone	varchar(100)	Altimetric zone label
location_id	integer	FK location_dt

4.4.5 Station Dimension (station_dt)

Attributes of air quality and weather monitoring stations. Each station links to its municipality through `municipality_dt` and to its coordinates through `location_dt`.

Column	Type	Description
station_id	integer	Primary key
name	varchar(100)	Station name
type	varchar(100)	Station type
municipality_id	integer	FK municipality_dt
location_id	integer	FK location_dt

4.4.6 Sensor Dimension (sensor_dt)

Individual sensor records, each tied to a station through **station_dt**. Includes measurement type, unit, and the operational date range.

Column	Type	Description
sensor_id	integer	Primary key
name	varchar(100)	Sensor name
type	varchar(100)	Measurement type (pollutant or variable)
uom	varchar(20)	Unit of measurement
date_start	date	First active date
date_stop	date	Last active date (NULL if still active)
station_id	integer	FK station_dt

4.4.7 Gate Dimension (gate_dt)

Area C access gates, identified by name and linked to their position through **location_dt**.

Column	Type	Description
gate_id	integer	Primary key
name	varchar(100)	Gate name
location_id	integer	FK location_dt

4.4.8 Vehicle Dimension (vehicle_dt)

Every distinct vehicle profile seen in the Area C data. Each row is a unique combination of service flag, vehicle type, fuel type, Euro standard, particulate filter, resident status, and Area C class.

Column	Type	Description
vehicle_id	integer	Primary key
service	varchar(100)	Service vehicle flag
type	varchar(100)	Vehicle type
fuel	varchar(100)	Fuel type
euro	varchar(100)	Euro emission standard
fap	varchar(100)	Particulate filter flag
allowed	varchar(100)	Exempt from Area C flag
resident	varchar(100)	Resident flag
category	varchar(100)	Vehicle category
class	varchar(100)	Area C vehicle class

4.4.9 Population Dimension (population_dt)

Annual resident population per municipality. Links a municipality to a reference year via **date_dt**.

Column	Type	Description
population_id	integer	Primary key
municipality_id	integer	FK municipality_dt
date_id	integer	FK date_dt
population	integer	Total resident population

5 ETL Flow

The ETL pipeline is written in Python, relying on standard libraries for database access and data manipulation. It is fully idempotent: every run drops and recreates the entire schema from scratch, so the result is always clean and reproducible. All datasets were downloaded manually and saved locally because not every source offers a direct download URL. Multi-year files were pre-filtered to 2023 to keep memory usage and runtime reasonable.

Three abstract base classes give the pipeline its structure.

```
@dataclass
class Pipeline(ABC):

    def run(self):
        self.extract()
        self.transform()
        self.load()

    @abstractmethod
    def extract(self): pass

    @abstractmethod
    def transform(self): pass

    @abstractmethod
    def load(self): pass

    @abstractmethod
    def get_df(self) -> pd.DataFrame: pass
```

Pipeline defines the extract-transform-load lifecycle. Concrete subclasses fill in the three steps for one target table each.

```
@dataclass
class Table(ABC):

    @abstractmethod
    def get_id(self, *args, **kwargs): pass
```

Once a pipeline finishes, its output **DataFrame** is wrapped in a **Table** subclass. The `get_id` method on each subclass contains the lookup logic for that table; downstream pipelines call it to resolve foreign keys.

```
@dataclass
class Source(ABC):

    @abstractmethod
    def extract(self) -> pd.DataFrame: pass
```

Source wraps a raw data file. Subclasses store format-specific details (delimiter, encoding) and cache the loaded **DataFrame** so that two pipelines reading the same file do not parse it twice.

5.1 Extraction Phase

Raw CSVs are loaded through typed **Source** subclasses. The resulting **DataFrame** goes straight to the transformation step.

5.2 Transformation Phase

Each pipeline takes the **Source** objects it needs and the **Table** objects of already-loaded dimensions. It outputs a cleaned **DataFrame** ready for loading. The transformation steps are:

- **Filtering and validation:** dropping invalid or out-of-scope records measurements with the missing-value sentinel, inactive sensors, entries outside 2023.
- **Schema harmonization:** renaming and selecting columns so that heterogeneous sources share one schema before any join or concatenation.
- **Code decoding:** replacing numeric or abbreviated codes with readable labels through lookup tables (e.g. urbanization levels, altimetric zones, Area C vehicle fields). Not every table needed this.
- **Temporal aggregation:** rolling data up to the target granularity. Sensor samples, for instance, are aggregated to daily level with mean, min, max, and standard deviation. Not every table needed it.
- **Foreign-key resolution:** matching each record to its dimension surrogate key. Where a plain join is not enough, the `get_id` method of the relevant **Table** handles the lookup, keeping resolution logic out of the pipeline code.
- **Referential integrity enforcement:** any row whose foreign key cannot be resolved is dropped, so no orphan records reach the database.

Because the snowflake schema creates dependencies between dimension tables, pipelines run in dependency order: each table is fully loaded before any downstream pipeline tries to resolve keys against it.

5.3 Load Phase

Each pipeline's `load` method writes its **DataFrame** to the target table. Dimensions are loaded first, then fact tables, respecting FK dependencies.

6 Analysis

Each analysis targets one of the business questions from Section 2, using a mix of Advanced SQL, Association Rule Mining, and Machine Learning Algorithms.

6.1 How can air quality data across municipalities be analyzed to support environmental and public health decision-making?

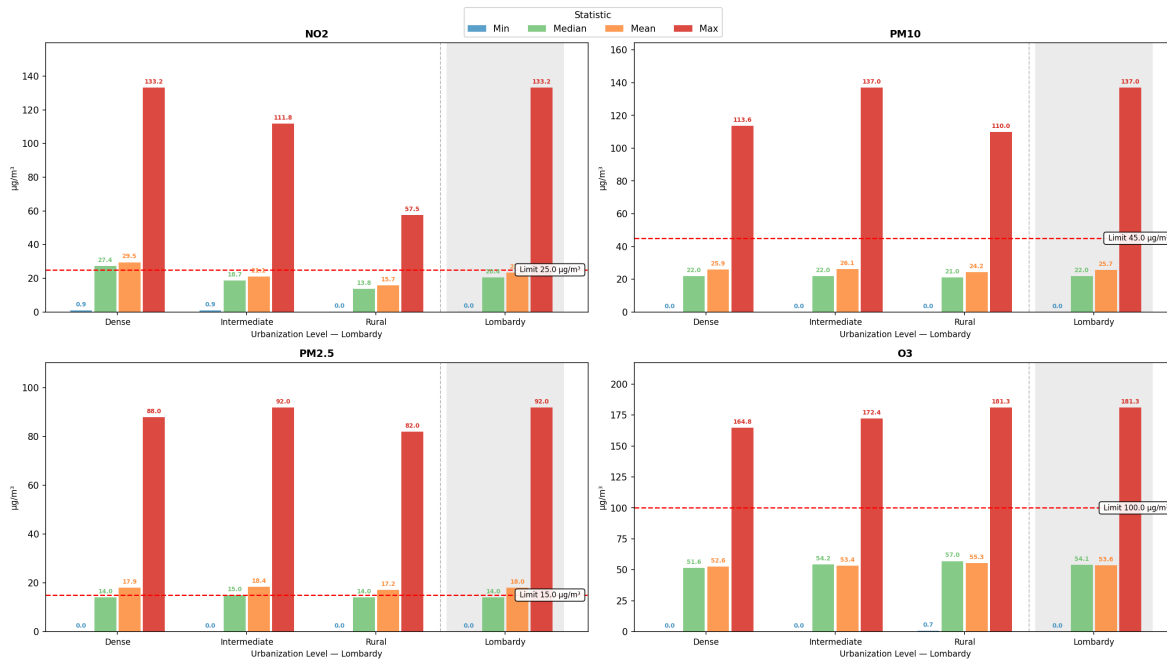
6.1.1 Which urbanization levels show the highest pollutant concentrations?

Four pollutants are considered: NO_2 , PM_{10} , $\text{PM}_{2.5}$, and O_3 . Each is compared against the WHO 2021 daily guideline values (source):

Pollutant	Concentration ($\mu\text{g}/\text{m}^3$)
NO_2	25
PM_{10}	45
$\text{PM}_{2.5}$	15
O_3	100

Table 1: Air Quality Thresholds

ISTAT classifies municipalities into three urbanization levels: **Dense**, **Intermediate**, and **Rural**. Statistics were computed per pollutant and urbanization level and compared to the Lombardy-wide aggregate.

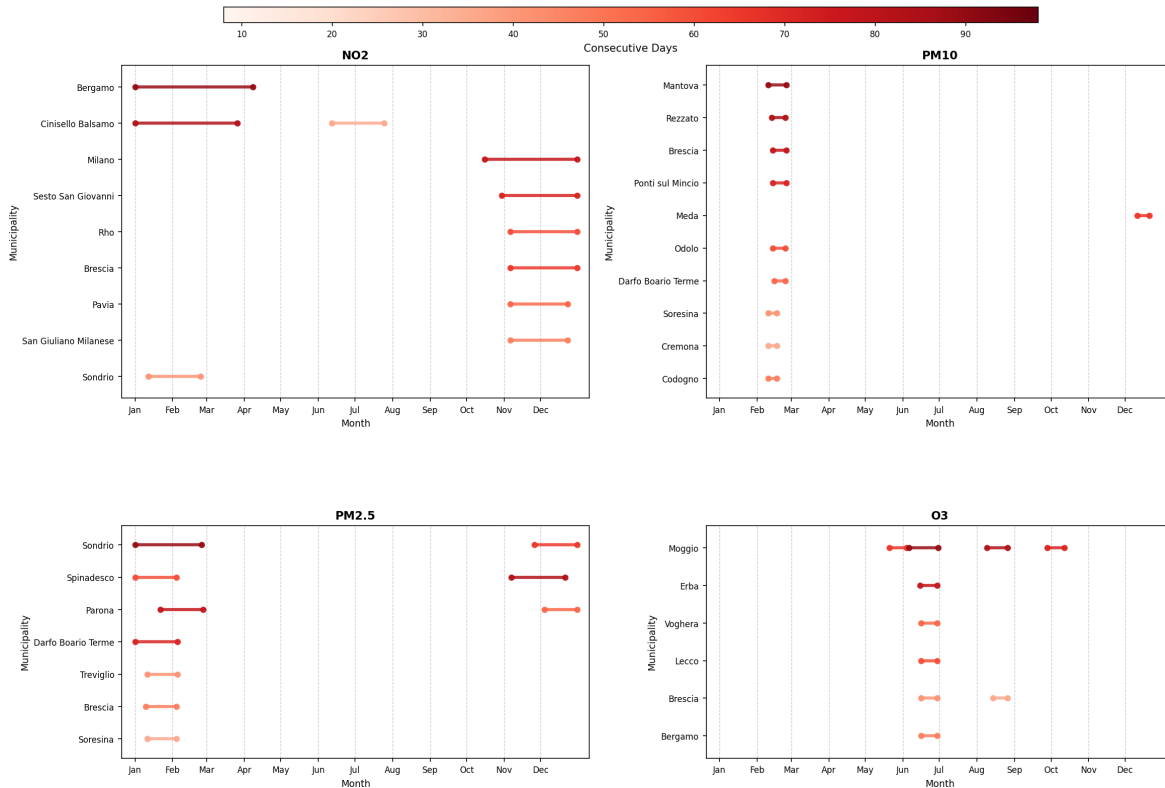


Particulate matter (PM_{10} and $\text{PM}_{2.5}$) is a **regional rather than urban problem**, lacking a clear urbanization gradient and even peaking in **Intermediate** areas. $\text{PM}_{2.5}$ is uniquely critical as it uniformly exceeds annual limits everywhere, demanding region-wide structural measures instead of local fixes. Pollutants patterns vary across urbanization levels. Dense areas are characterized by

higher NO_2 concentrations, rural areas by elevated O_3 levels linked to reduced NO titration, while intermediate zones show the highest PM_{10} and $\text{PM}_{2.5}$ concentrations.

6.1.2 Are there municipalities that repeatedly experience multi-day episodes of pollution above health-based thresholds?

To find municipalities with recurring sustained exceedances, **critical pollution windows** were defined: consecutive days on which a municipality's daily average stays above the threshold for a given pollutant. The charts show the top 10 longest consecutive exceedance episodes per pollutant; darker red means longer duration.



Exceedances cluster in predictable calendar windows.

Two cases break the pattern. **Cinisello Balsamo** registers an NO_2 episode in mid-summer, when traffic is typically low. That is unusual and suggests a local non-traffic source. **Moggio** shows O_3 exceedances from May all the way to October far longer than anywhere else.

Episode duration matters as much as frequency. A week-long exceedance is a different health risk from a one-day spike, and any risk assessment should weight it accordingly.

6.1.3 Can municipalities be grouped based on their pollution profile?

K-Means Clustering was run on daily pollution profiles to group municipalities. The feature matrix contains the daily average concentration of NO_2 , PM_{10} , $\text{PM}_{2.5}$, and O_3 effectively a 365-day time series per pollutant for each municipality. Municipalities with less than 50% daily coverage across the full matrix were dropped. Missing values in the remaining rows were filled with the **column median**, that is, the region-wide daily median for that pollutant on that day. A stricter filter demanding all

four pollutants would have left too few municipalities to cluster usefully. Features were scaled with a **Robust Scaler** to dampen outlier effects.

$k = 3$ was chosen after inspecting the Elbow Chart, Silhouette Score, and Davies-Bouldin Index.

For each cluster, weekly averages were plotted against guideline thresholds. A **Cluster Pollution Index**, the mean per-pollutant concentration normalized by its threshold, was computed to rank clusters from dirtiest to cleanest.

Cluster characterization

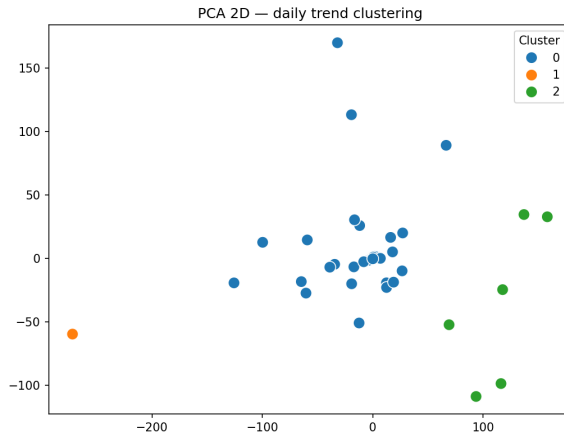


Figure 4: 2D PCA projection

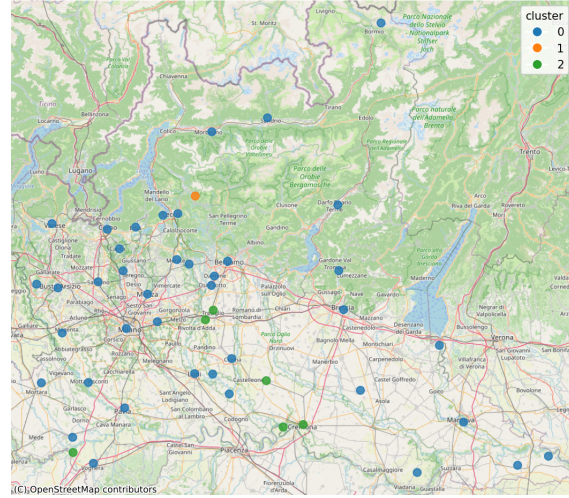


Figure 5: Cluster location map

Cluster 0 contains most of the monitored municipalities. **Cluster 2** picks up a smaller, more polluted subset. **Cluster 1** consists of a **single municipality**: its profile is so different from everything else that it ends up alone. It is best read as an outlier, not a representative group.

Weekly Seasonal trends



Figure 6: Weekly pollutants trend.

The seasonal shape is similar everywhere, primary pollutants (NO_2 , PM_{10} , $\text{PM}_{2.5}$) peak in winter and drop through spring and summer; O_3 does the opposite, peaking in summer.

Cluster 2 is the more polluted group. Its primary-pollutant concentrations are the highest year-round, and in winter both $\text{PM}_{2.5}$ and NO_2 regularly breach guideline levels. **Cluster 0** sits near the regional average. **Cluster 1**, the lone municipality, has readings extreme enough to form its own cluster, confirming it is a statistical outlier.

Cluster Pollution Index

Cluster	Pollution Index
2	0.887
0	0.779
1	0.471

Cluster Pollution Index.

Cluster 2 has the highest index (0.887). **Cluster 0** comes next at 0.779. **Cluster 1** is well below at 0.471.

6.1.4 Which municipalities are most polluted with particulate matter?

The top 5 most particulate-polluted municipalities were extracted month by month for PM_{2.5} and PM₁₀. Appearances were then broken down by urbanization level and compared to the baseline distribution of monitored municipalities (23% Dense, 56% Intermediate, 21% Rural).

Top 5 municipalities per month PM_{2.5}

Month	1 st	2 nd	3 rd	4 th	5 th
January	Spinadesco	Soresina	Darfo Boario Terme	Treviglio	Cremona
February	Ponti sul Mincio	Mantova	Soresina	Spinadesco	Brescia
March	Soresina	Casirate d'Adda	Spinadesco	Bergamo	Treviglio
April	Spinadesco	Casirate d'Adda	Dalmine	Soresina	Treviglio
May	Bergamo	Spinadesco	Casirate d'Adda	Treviglio	Pavia
June	Dalmine	Treviglio	Casirate d'Adda	Spinadesco	Saronno
July	Casirate d'Adda	Cornale e Bastida	Soresina	Cremona	Spinadesco
August	Spinadesco	Soresina	Casirate d'Adda	Cremona	Cornale e Bastida
September	Casirate d'Adda	Soresina	Spinadesco	Cremona	Cornale e Bastida
October	Soresina	Cornale e Bastida	Spinadesco	Casirate d'Adda	Treviglio
November	Spinadesco	Soresina	Cremona	Cornale e Bastida	Casirate d'Adda
December	Soresina	Spinadesco	Cremona	Monza	Casirate d'Adda

PM_{2.5} top-5 appearances.

Urbanization	Appearances	Share (%)
Dense	12	20.0
Intermediate	31	51.7
Rural	17	28.3

PM_{2.5} top-5 appearances by urbanization level.

Top 5 municipalities per month PM₁₀

Month	1 st	2 nd	3 rd	4 th	5 th
January	Darfo Boario Terme	Rezzato	Soresina	Codogno	Meda
February	Rezzato	Mantova	Ponti sul Mincio	Brescia	Soresina
March	Rezzato	Soresina	Codogno	Brescia	Casirate d'Adda
April	Spinadesco	Soresina	Casirate d'Adda	Dalmine	Darfo Boario Terme
May	Spinadesco	Soresina	Codogno	Casirate d'Adda	Brescia
June	Borgocarbonara	Spinadesco	Codogno	Rezzato	Soresina
July	Borgocarbonara	Spinadesco	Soresina	Casirate d'Adda	Codogno
August	Soresina	Codogno	Spinadesco	Crema	Borgocarbonara
September	Soresina	Spinadesco	Casirate d'Adda	Cremona	Crema
October	Soresina	Codogno	Rezzato	Crema	Spinadesco
November	Soresina	Rezzato	Spinadesco	Codogno	Cremona
December	Rezzato	Meda	Soresina	Mantova	Cremona

PM₁₀ top-5 appearances.

Urbanization	Appearances	Share (%)
Dense	8	13.3
Intermediate	41	68.3
Rural	11	18.3

PM₁₀ top-5 appearances by urbanization level.

Dense municipalities are **under-represented** relative to their share of the monitoring network, Intermediate and Rural areas are **over-represented** for both pollutants.

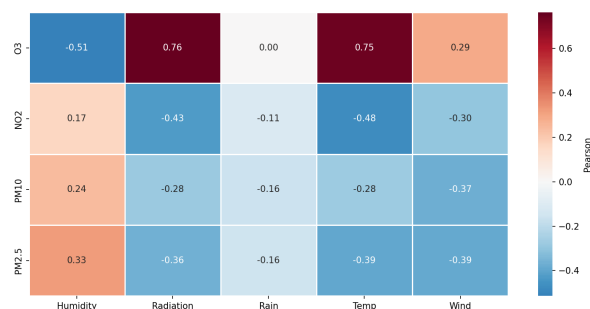
The gap is widest for PM₁₀: Intermediate municipalities make up 68.3% of appearances versus a 56% baseline, while Dense areas drop to 13.3% against 23%. One plausible reading is that urban traffic restrictions and centralized heating regulations are doing their job in large cities, but peripheral industrial and agricultural zones remain outside current policy reach.

For PM_{2.5}, Rural municipalities appear in 28.3% of slots versus a 21% baseline. That pattern fits biomass burning and agricultural emissions sources that urban air quality frameworks barely touch.

6.2 How do weather variables relate to pollutant concentrations across municipalities?

6.2.1 Which weather variables are most strongly correlated with pollutant concentrations?

Pearson correlation coefficients were computed between daily weather variables (temperature, humidity, solar radiation, wind speed, precipitation) and daily pollutant concentrations at municipality level.



O₃ correlates strongly with radiation and temperature (0.76 and 0.75), no surprise given its photochemical origin. The primary pollutants (NO₂, PM₁₀, PM_{2.5}) go the other way, showing negative correlations with the same variables: they accumulate under cold, stagnant, low-radiation conditions.

Humidity is the only variable that flips sign across pollutant groups: positive for particulates (up to 0.33 for PM_{2.5}), negative for O₃ (−0.51). At daily resolution, humidity apparently helps PM accumulate rather than washing it out. **Rain**, on the other hand, is nearly uncorrelated with everything (0.00 to −0.16). Precipitation by itself is a poor predictor of air quality.

6.2.2 Which combinations of weather conditions frequently co-occur with pollution levels?

Association Rule Mining was used to find which weather combinations tend to appear together with specific pollution levels, picking up non-linear and multi-variable patterns that Pearson coefficients miss.

Both weather and pollutant variables were discretized into three ordinal bins (LOW, MID, HIGH) using **per-municipality tertiles**, so thresholds reflect local conditions.

Apriori was run on the resulting binary matrix with minimum support 0.5% and minimum confidence 20%; only rules with lift ≥ 1.20 and a weather-only antecedent mapping to a pollutant-only consequent were kept.

Antecedent	Consequent	Support	Confidence	Lift
RADIATION_MID + RAIN_LOW + TEMP_MID + WIND_LOW	O3_MID + PM10_HIGH	0.64%	37.1%	7.37
RAIN_LOW + TEMP_HIGH + WIND_LOW	O3_HIGH + PM10_HIGH	0.75%	23.1%	6.90
HUMIDITY_MID + RADIATION_MID + RAIN_LOW + WIND_LOW	NO2_HIGH + O3_MID + PM10_HIGH	0.57%	21.7%	6.81
HUMIDITY_MID + RAIN_LOW + TEMP_MID + WIND_LOW	O3_MID + PM10_HIGH	0.62%	33.8%	6.72
HUMIDITY_MID + RADIATION_MID + WIND_LOW	NO2_HIGH + O3_MID + PM10_HIGH	0.59%	20.5%	6.45
HUMIDITY_MID + RADIATION_MID + RAIN_LOW + TEMP_MID	NO2_HIGH + O3_MID + PM10_HIGH	0.50%	20.5%	6.43
HUMIDITY_HIGH + RADIATION_LOW + RAIN_HIGH + TEMP_MID	O3_LOW + PM10_LOW	0.65%	23.6%	6.30
HUMIDITY_MID + RADIATION_HIGH + RAIN_HIGH + WIND_HIGH	NO2_LOW + O3_HIGH + PM10_LOW	0.70%	46.7%	6.25
HUMIDITY_MID + TEMP_MID + WIND_LOW	O3_MID + PM10_HIGH	0.68%	31.4%	6.25
RADIATION_MID + TEMP_MID + WIND_LOW	O3_MID + PM10_HIGH	0.69%	31.0%	6.16
HUMIDITY_MID + RADIATION_HIGH + RAIN_HIGH + TEMP_HIGH + WIND_HIGH	NO2_LOW + O3_HIGH + PM10_LOW	0.50%	45.8%	6.13
HUMIDITY_MID + RADIATION_MID + RAIN_LOW + TEMP_MID	O3_MID + PM10_HIGH	0.75%	30.6%	6.08
HUMIDITY_MID + RADIATION_MID + RAIN_LOW + WIND_LOW	O3_MID + PM10_HIGH	0.79%	30.4%	6.05
RADIATION_LOW + RAIN_HIGH + TEMP_MID	O3_LOW + PM10_LOW	0.69%	22.3%	5.96
RADIATION_MID + RAIN_LOW + TEMP_LOW + WIND_MID	NO2_HIGH + O3_MID	0.53%	35.5%	5.93
HUMIDITY_MID + RADIATION_MID + WIND_LOW	O3_MID + PM10_HIGH	0.85%	29.6%	5.88
HUMIDITY_MID + RADIATION_HIGH + RAIN_HIGH	NO2_LOW + O3_HIGH + PM10_LOW	1.00%	42.9%	5.74
HUMIDITY_HIGH + RADIATION_LOW + TEMP_MID	O3_LOW + PM10_LOW	0.73%	21.3%	5.70
HUMIDITY_MID + RADIATION_HIGH + RAIN_HIGH + TEMP_HIGH	NO2_LOW + O3_HIGH + PM10_LOW	0.75%	42.5%	5.68
RADIATION_HIGH + RAIN_HIGH + TEMP_HIGH + WIND_HIGH	NO2_LOW + O3_HIGH + PM10_LOW	0.67%	42.0%	5.62

Top 20 rules ranked by lift (Weather $\rightarrow$ Pollutants).

Two meteorological patterns emerge from the rules, each with its own pollution signature.

The first and most common is **atmospheric stagnation**: low wind, little rain, moderate temperature. The top-ranked rule RADIATION_MID + RAIN_LOW + TEMP_MID + WIND_LOW $\rightarrow$ O3_MID + PM10_HIGH reaches a lift of 7.37, and the same WIND_LOW + RAIN_LOW core reappears across most top rules with lifts in the 6.05–7.37 range, making stagnation the single best meteorological predictor of pollution peaks.

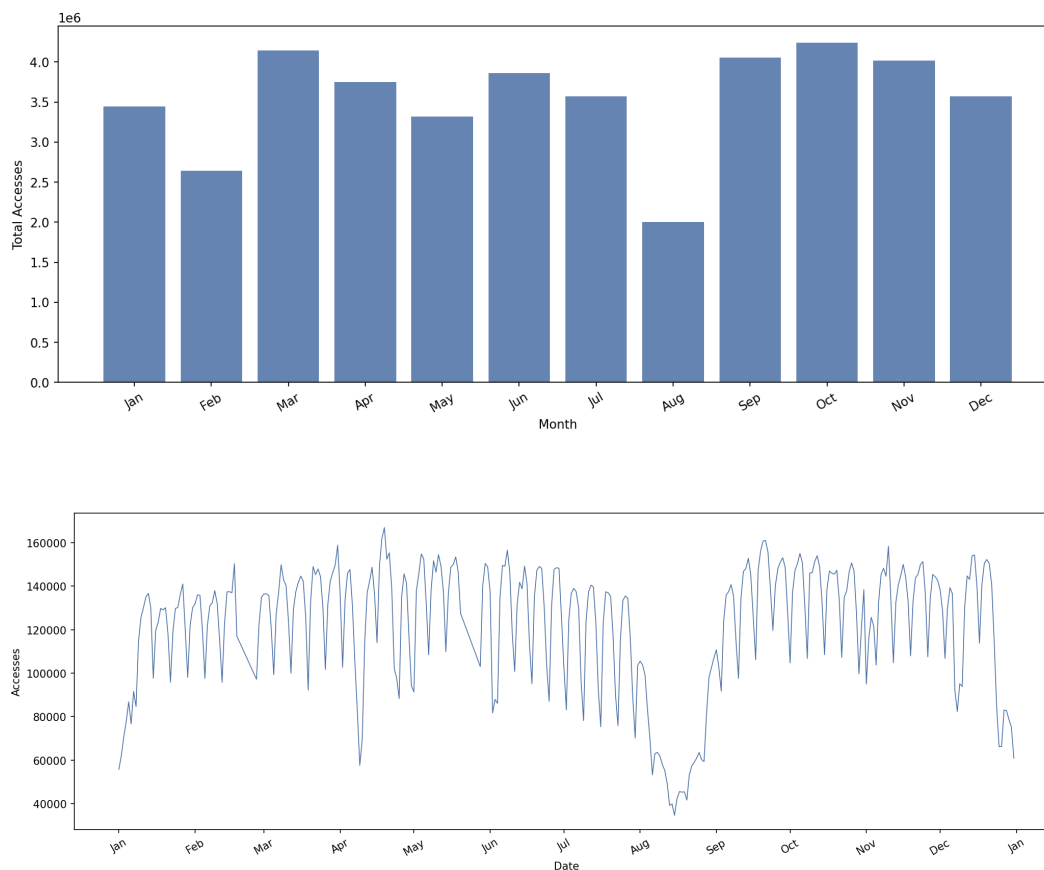
The second is a **decoupling pattern** where high radiation coincides with rain and wind. Rule HUMIDITY_MID + RADIATION_HIGH + RAIN_HIGH + WIND_HIGH $\rightarrow$ NO2_LOW + O3_HIGH + PM10_LOW (confidence 46.7%, lift 6.25) captures it best: NO₂ and PM₁₀ drop to low levels (wet scavenging clears them out), but O₃ stays high because photochemistry keeps running as long as there is enough sunlight.

6.3 How can vehicle access data be analyzed to model patterns, forecast trends, and detect anomalies in support of mobility planning?

6.3.1 What is the composition of the fleet of vehicles accesses and how is it distributed?

Several queries were run to extract the access trend and fleet breakdown by fuel type and Euro standard. The results were plotted to spot dominant categories and seasonal patterns.

Access trend



Accesses peak in **October** and bottom out in **August**. The daily series makes it clear that day-to-day swings come from a rigid **weekly cycle**, not from any gradual seasonal drift.

Fleet composition by fuel type

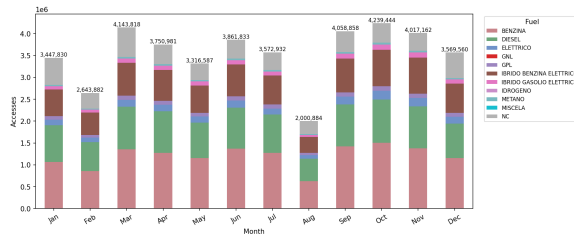


Figure 7: Monthly fleet composition by fuel type.

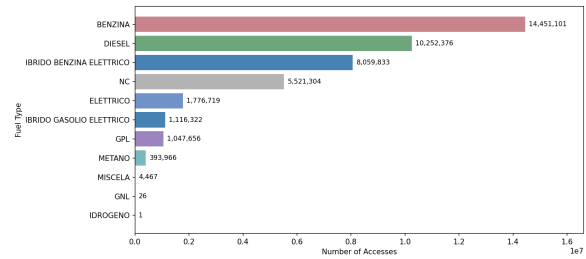


Figure 8: Total accesses by fuel type.

Benzina and Diesel together make up about 58% of accesses, but **Ibrido Benzina Elettrico** already accounts for 19%, pushing total low-emission share to roughly 23%. Alternative fuels are marginal (<3.5%). Hydrogen: one single access all year. The **NC** category (13%) is large enough to be a real problem for any emission estimate built on fleet composition. Month-to-month, the mix barely changes.

Fleet composition by Euro standard

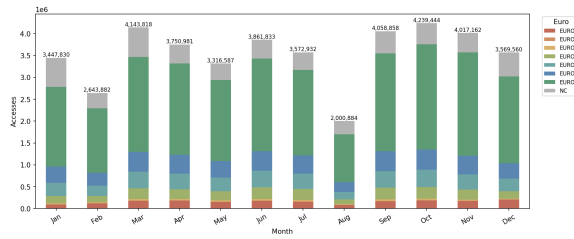


Figure 9: Monthly fleet composition by Euro standard.

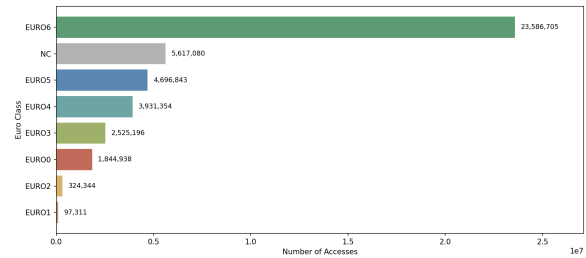


Figure 10: Total accesses by Euro standard.

Euro 6 vehicles make up over 55% of accesses, a direct consequence of progressive urban access bans pushing fleet renewal. The **Euro 0** share (about 4%) is actually higher than Euro 1 and Euro 2 combined, most likely motorcycles and historic vehicles benefiting from exemptions, not genuine non-compliance.

6.3.2 Can daily vehicle accesses be predicted?

A **Random Forest Regressor** was trained to predict daily accesses. Features includes lagged traffic (1, 2, 3, and 7 days back), a 7-day rolling mean and standard deviation, day of week, month, a weekend flag, a holiday indicator, and a bridge-day indicator. Hyperparameters were tuned by **Cross Validation** over an 11-fold **Time Series Split**, so every test fold comes strictly after its training data.

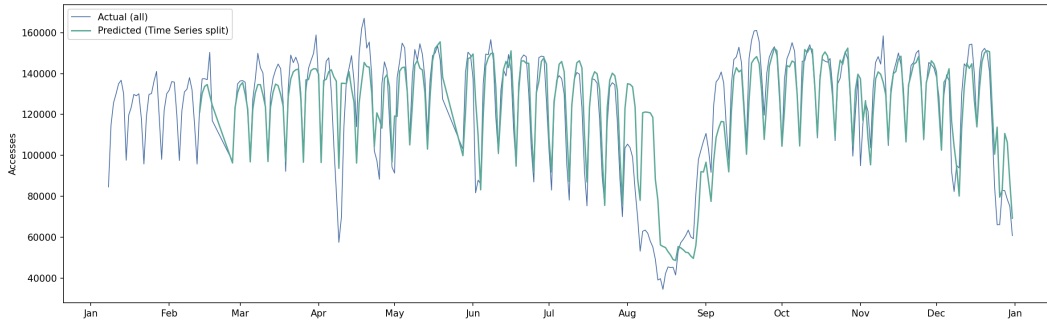


Figure 11: Actual vs. predicted daily traffic volumes.

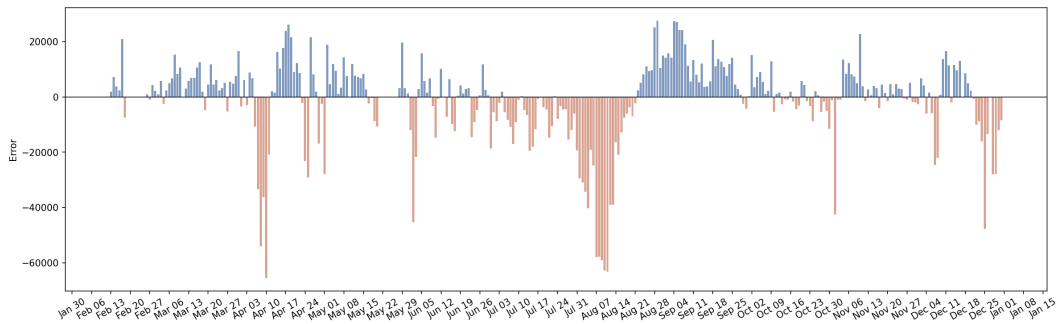


Figure 12: Residuals of the daily traffic volume predictions.

Metric	Value
MAE	10,666 accesses/day
RMSE	15,716 accesses/day
MAPE	11.2%
R^2	0.724

The features capture about 72% of daily variance. The missing 28% is presumably related to other feature not mapped like weather, big public events, and transport strikes. The model over-predicts because its lag features carry forward normal working-day volumes and cannot anticipate the sudden drop. Holiday and bridge indicator were added to compensate but is not sufficient. The mirror problem shows up on post-holiday return days where traffic rebounds sharply, producing the largest under-predictions.

6.3.3 Which days show anomalous traffic volumes?

An **Isolation Forest** (200 estimators, 5% contamination) was fitted on daily traffic totals.

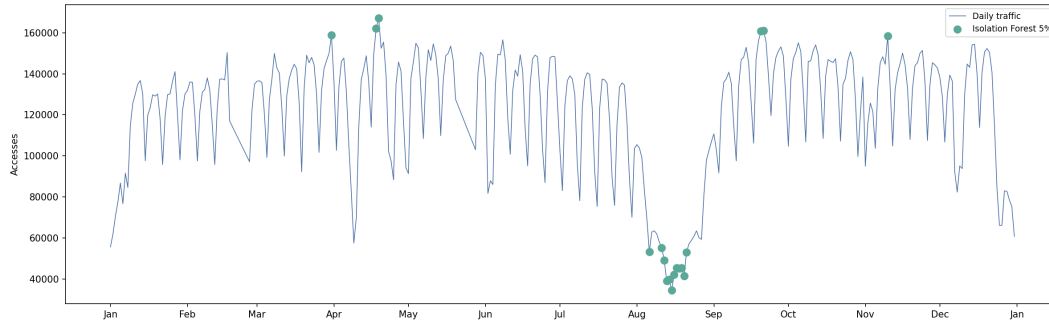


Figure 13: Isolation Forest anomaly detection results.

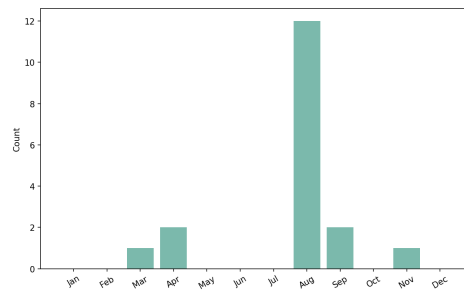


Figure 14: Monthly distribution of detected anomalies.

18 days were flagged, splitting neatly into two groups. The **low-traffic anomalies** cluster around Ferragosto: Italy's main summer shutdown pushes volumes well below baseline. The **high-traffic anomalies** land on predictable calendar transitions: the days around Easter, the end-of-summer rebound, and Fridays before long weekends. Every flagged day has an obvious calendar explanation. Area C traffic is, in short, very regular once you account for a handful of known events.

6.4 Is there a measurable association between human activity intensity and air pollutant levels?

6.4.1 Do high traffic days co-occur with elevated pollutant levels?

The activity variable here is **Area C daily traffic volume**, paired with daily concentrations of NO₂, O₃, PM₁₀, and PM_{2.5}. Because the traffic data covers only one zone, **this analysis is limited to the municipality of Milan**. Both traffic and pollutant values were binned into three levels (LOW, MID, HIGH) using tertiles over the Milan daily series. Apriori was run with minimum support 0.5% and minimum confidence 20%. Only rules with a traffic-only antecedent, pollutant-only consequent, and lift ≥ 1.0 were kept.

Antecedent	Consequent	Support	Confidence	Lift
TRAFFIC_LOW	NO2_LOW + O3_HIGH + PM10_LOW	8.26%	24.8%	1.89
TRAFFIC_LOW	NO2_LOW + O3_HIGH	15.67%	47.0%	1.83
TRAFFIC_LOW	NO2_LOW + O3_HIGH + PM25_MID	7.98%	23.9%	1.83
TRAFFIC_LOW	NO2_LOW + PM25_MID	8.83%	26.5%	1.82
TRAFFIC_LOW	NO2_LOW + O3_HIGH + PM25_LOW	7.41%	22.2%	1.81
TRAFFIC_LOW	O3_HIGH + PM10_LOW + PM25_LOW	6.84%	20.5%	1.80
TRAFFIC_LOW	O3_HIGH + PM10_LOW	8.55%	25.6%	1.80
TRAFFIC_LOW	NO2_LOW + PM10_LOW	11.40%	34.2%	1.79
TRAFFIC_LOW	NO2_LOW + PM10_LOW + PM25_LOW	9.40%	28.2%	1.77
TRAFFIC_LOW	NO2_LOW	19.37%	58.1%	1.74
TRAFFIC_LOW	O3_HIGH + PM25_LOW	7.69%	23.1%	1.72
TRAFFIC_HIGH	NO2_MID + PM25_MID	6.84%	20.5%	1.71
TRAFFIC_LOW	NO2_LOW + PM10_MID	6.84%	20.5%	1.71
TRAFFIC_LOW	NO2_LOW + PM25_LOW	10.26%	30.8%	1.66
TRAFFIC_LOW	O3_HIGH	17.09%	51.3%	1.54
TRAFFIC_LOW	O3_HIGH + PM25_MID	8.26%	24.8%	1.47
TRAFFIC_MID	NO2_HIGH + O3_LOW + PM25_HIGH	8.83%	26.5%	1.45
TRAFFIC_MID	NO2_HIGH + O3_LOW	11.11%	33.3%	1.44
TRAFFIC_MID	NO2_HIGH + O3_LOW + PM10_HIGH	8.26%	24.8%	1.43
TRAFFIC_MID	NO2_HIGH + O3_LOW + PM10_HIGH + PM25_HIGH	7.98%	23.9%	1.42

Top 20 rules ranked by lift (Traffic $\rightarrow$ Pollutants).

Three traffic patterns emerge from the rules, but none reflects a direct causal link with emissions.

The TRAFFIC_LOW rules dominate the top of the table. The strongest, TRAFFIC_LOW $\rightarrow$ NO2_LOW + O3_HIGH + PM10_LOW (lift 1.89), and the rest of the TRAFFIC_LOW group (lifts 1.54–1.83) all point to the same NO2_LOW + O3_HIGH signature. This is **seasonal confounding**, low traffic and high ozone co-occur because August kills commuter traffic exactly when summer meteorology pushes O₃ up. A calendar effect, not a causal link.

The TRAFFIC_MID pattern captures the opposite season. Rule TRAFFIC_MID $\rightarrow$ NO2_HIGH + O3_LOW + PM25_HIGH (lift 1.45) anchors a tight group (lifts 1.42–1.45) where intermediate volumes pair with elevated NO₂ and PM_{2.5}. This matches autumn and winter workdays, when **atmospheric stagnation**.

The TRAFFIC_HIGH pattern yields a single rule, TRAFFIC_HIGH $\rightarrow$ NO2_MID + PM25_MID (lift 1.71): peak volumes produce no distinctive pollution signature beyond what the season already explains.

The overall maximum lift of 1.89 is low by association-rule standards. Two structural limitations apply. First, **geography**, traffic and pollution data are both restricted to Milan, so nothing here

generalizes to the rest of Lombardy. Second, **instrument mismatch**, Area C monitors a single corridor into the city centre and captures only a fraction of total urban mobility, making it a weak proxy for actual emission intensity. A more informative analysis would need region-wide counts from motorways, provincial roads, and other cities.

6.4.2 Does high tourism pressure co-occur with higher pollution levels?

Here the activity variable is the **monthly tourism pressure index**, compared against daily concentrations of NO₂, O₃, PM₁₀, and PM_{2.5}. Tourism data is natively monthly, so both activity and pollution were discretized into LOW, MID, HIGH using **per-municipality monthly tertiles**. Apriori ran with minimum support 0.5% and minimum confidence 20%, keeping only rules with lift ≥ 1.0 , tourism-only antecedent, and pollutant-only consequent.

Antecedent	Consequent	Support	Confidence	Lift
TOURISM_LOW	O3_LOW + PM10_HIGH + PM25_HIGH	6.87%	20.5%	2.65
TOURISM_LOW	NO2_HIGH + PM10_HIGH + PM25_HIGH	8.93%	26.7%	2.63
TOURISM_LOW	PM10_HIGH + PM25_HIGH	9.97%	29.7%	2.55
TOURISM_LOW	O3_LOW + PM25_HIGH	7.22%	21.5%	2.51
TOURISM_LOW	NO2_HIGH + O3_LOW + PM25_HIGH	6.87%	20.5%	2.49
TOURISM_LOW	NO2_HIGH + O3_LOW + PM10_HIGH	10.48%	31.3%	2.40
TOURISM_LOW	O3_LOW + PM10_HIGH	11.17%	33.3%	2.40
TOURISM_LOW	NO2_HIGH + PM25_HIGH	9.62%	28.7%	2.39
TOURISM_LOW	NO2_HIGH + PM10_HIGH	15.81%	47.2%	2.33
TOURISM_LOW	PM25_HIGH	10.82%	32.3%	2.29
TOURISM_LOW	NO2_HIGH + O3_LOW	14.09%	42.1%	2.13
TOURISM_LOW	PM10_HIGH	17.87%	53.3%	2.07
TOURISM_LOW	O3_LOW	15.64%	46.7%	2.01
TOURISM_LOW	NO2_HIGH	22.34%	66.7%	2.00
TOURISM_HIGH	NO2_LOW + O3_HIGH	11.51%	34.5%	1.72
TOURISM_HIGH	O3_HIGH	12.37%	37.1%	1.64
TOURISM_HIGH	NO2_LOW	18.21%	54.6%	1.63
TOURISM_HIGH	NO2_LOW + PM10_LOW	10.14%	30.4%	1.58
TOURISM_HIGH	PM25_MID	7.22%	21.6%	1.56
TOURISM_HIGH	NO2_MID + PM10_MID	6.87%	20.6%	1.48

Top 20 rules ranked by lift (Tourism $\rightarrow$ Pollutants).

Two tourism patterns emerge from the rules, but both reduce to seasonal weather rather than visitor activity.

The TOURISM_LOW rules fill 14 of the top 20 slots. The strongest, TOURISM_LOW $\rightarrow$ O3_LOW + PM10_HIGH + PM25_HIGH (lift 2.65), anchors a tight group (lifts 2.00–2.65) that maps onto the classic winter fingerprint: NO2_HIGH, PM10_HIGH, PM25_HIGH, O3_LOW. The single highest-confidence rule, TOURISM_LOW $\rightarrow$ NO2_HIGH (66.7%), confirms the picture. This is **thermal inversion** in winter, not an effect of fewer tourists.

The TOURISM_HIGH pattern captures the opposite season. Rule TOURISM_HIGH $\rightarrow$ NO2_LOW + O3_HIGH (lift 1.72) anchors a smaller group (6 rules, lifts 1.48–1.72) pairing peak visitor months with the photochemical signature of summer: low NO₂ and high O₃, exactly what high radiation and better ventilation produce. Nothing here points to tourism itself improving air quality.

Two structural limitations weaken these patterns. First, **temporal granularity**, tourism is reported monthly, so every day within a month inherits the same TOURISM_LOW/MID/HIGH label. Daily variability in the antecedent is gone, and the rules effectively reduce to a comparison between calendar months. The high lifts (up to 2.65) reflect the strong seasonal contrast in pollution, not a sharp dependence on

visitor numbers. Second, the monthly index conflates **season and tourism intensity**: low-tourism months are also winter months, so the antecedent cannot be cleanly separated from the meteorology it correlates with.

Tourism intensity therefore acts as a **calendar proxy**. No rule connects high visitor numbers to pollution levels above what the season already produces, and municipality-scale air quality is driven by large-scale meteorology, not visitor counts. A finer-grained tourism signal (daily arrivals, overnight stays) would be needed to test whether visitor pressure has any residual effect once seasonality is controlled for.

7 Generative artificial intelligence tools used

AI tools like GitHub Copilot and Google Gemini were used during the project for:

- support writing and debugging python code
- support writing documentation, checking text and fixing grammar